Retrieval-Augmented Language Models for Earnings Forecasting

FICTIONAL paper created for software demonstration only. Authors, data and results are invented.

Abstract

We study whether retrieval-augmented language models can improve analyst-style forecasts of quarterly earnings surprises. We propose EarnRAG, a method that retrieves relevant passages from prior filings and earnings-call transcripts and conditions a language model on them. On a dataset of 18,400 quarterly reports from 1,210 U.S. firms (2012-2024), EarnRAG improves directional accuracy of earnings-surprise prediction from 54.1% to 61.7% over a strong gradient-boosting baseline. However, gains concentrate in large-cap firms and vanish during the 2020 volatility regime, which we discuss as a key limitation.

1. Introduction

Forecasting whether a firm will beat or miss consensus earnings is a central task for investors. Traditional models rely on structured features such as past surprises and analyst dispersion, but ignore the rich text in filings. Large language models can read text, yet they hallucinate figures when asked to recall facts. We hypothesize that retrieving evidence from a firm's own prior documents reduces hallucination and improves forecasts.

2. Method

EarnRAG has three stages. First, each filing is split into 512-token passages and embedded with a dense retriever. Second, for each forecast we retrieve the top 8 passages from the firm's previous four quarters using hybrid lexical and dense search. Third, a language model is prompted with the retrieved passages and structured features and outputs a probability that the firm will beat consensus. We fine-tune the final classifier head with a cross-entropy loss and calibrate probabilities with temperature scaling.

3. Experiments

Data. We use 18,400 firm-quarter observations from 1,210 firms between 2012 and 2024, split chronologically: 2012-2020 for training, 2021 for validation and 2022-2024 for testing. Baselines include logistic regression on structured features, gradient boosting (XGBoost) and a zero-shot language model without retrieval.

Model	Directional accuracy	AUC	Brier score
Logistic regression	52.3%	0.538	0.261
Gradient boosting	54.1%	0.571	0.253
Zero-shot LLM (no retrieval)	55.0%	0.580	0.255
EarnRAG (ours)	61.7%	0.659	0.236

Results. EarnRAG improves directional accuracy by 7.6 percentage points over gradient boosting ($p < 0.01$, paired bootstrap). Removing retrieval drops accuracy to 55.0%, and removing structured features drops it to 58.9%, showing that both components contribute. Calibration also improves, with Brier score falling from 0.253 to 0.236.

4. Limitations

Gains are concentrated in large-cap firms with rich disclosure: accuracy is 64.8% for the top size tercile but only 56.2% for the bottom tercile. During the 2020 volatility regime, EarnRAG's accuracy fell to 52.0%, no better than chance. Our backtest ignores transaction costs and we did not test whether the signal survives after publication. Finally, evaluation uses a single market (U.S.) and results may not transfer to other disclosure regimes.

5. Conclusion

Retrieval grounds language models in firm-specific evidence and yields meaningful gains in earnings-surprise prediction, but the benefit is uneven across firm size and market regimes. Future work will study robustness, cost-aware trading evaluation and multilingual filings.